IGCSE 0607 Practice Questions — E4.5

Parallel lines

Total Questions: 9 — **Total Marks:** 44 Marks

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1 Find the equation of the line that is parallel to $y = -2x + 7$ and passes through the point $(3, 4)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

2 Line L has the equation $4x + 2y = 9$.
Find the equation of the line parallel to L that passes through the point $(3, -1)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

3 Line L has the equation $y = 6 - 3x$.

Find the equation of the line parallel to L that passes through the point $(-2, 5)$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

4 The points A and B have coordinates $(-3, 5)$ and $(5, -1)$ respectively.
Line L is parallel to the line segment AB and passes through the point $(4, -2)$.

(a) Find the gradient of the line segment AB.

..... **[2]**

(b) Find the equation of line L.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

5 Line M has the equation $5x - 2y = 15$.
Point P has coordinates $(-4, 1)$.
Find the equation of the line parallel to M that passes through P.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

6 A is the point $(-3, 2)$ and B is the point $(5, 6)$.

(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Line L is parallel to AB and passes through the point $(4, -1)$.
Find the equation of line L, giving your answer in the form $ax + by = c$, where a, b and c are integers.

..... **[3]**

7 Points $A(-1, 5)$ and $B(3, -3)$ lie on line L_1.
Line L_2 is parallel to L_1 and passes through the point $C(2, 7)$.
Line L_2 crosses the x-axis at D and the y-axis at E.

(a) Find the equation of line L_2.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

(b) Calculate the exact area of triangle ODE, where O is the origin $(0, 0)$.

..... $units^2$ **[3]**

8 Line L_1 has the equation $3x - 4y = 12$.
Line L_2 is parallel to L_1.

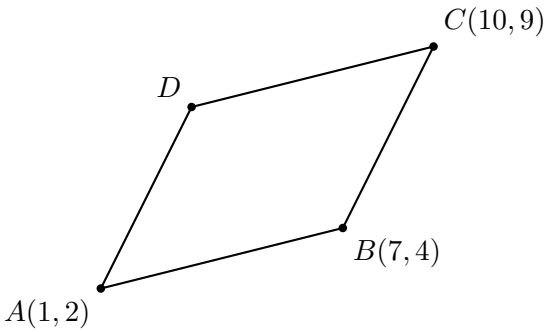
Line L_2 crosses the x-axis at point A and the y-axis at point B.
The length of the line segment AB is 15.

Find the two possible equations of L_2.
Give your answers in the form $y = mx + c$.

.....

..... **[7]**

9 The points $A(1, 2)$, $B(7, 4)$, and $C(10, 9)$ are three vertices of a parallelogram $ABCD$.
Line CD is parallel to line AB and line AD is parallel to line BC.



Not to scale

(a) Find the equation of the line CD.

..... **[3]**

(b) Find the equation of the line AD.

..... **[3]**

(c) Work out the coordinates of the vertex D.

(..... , ...) **[3]**